

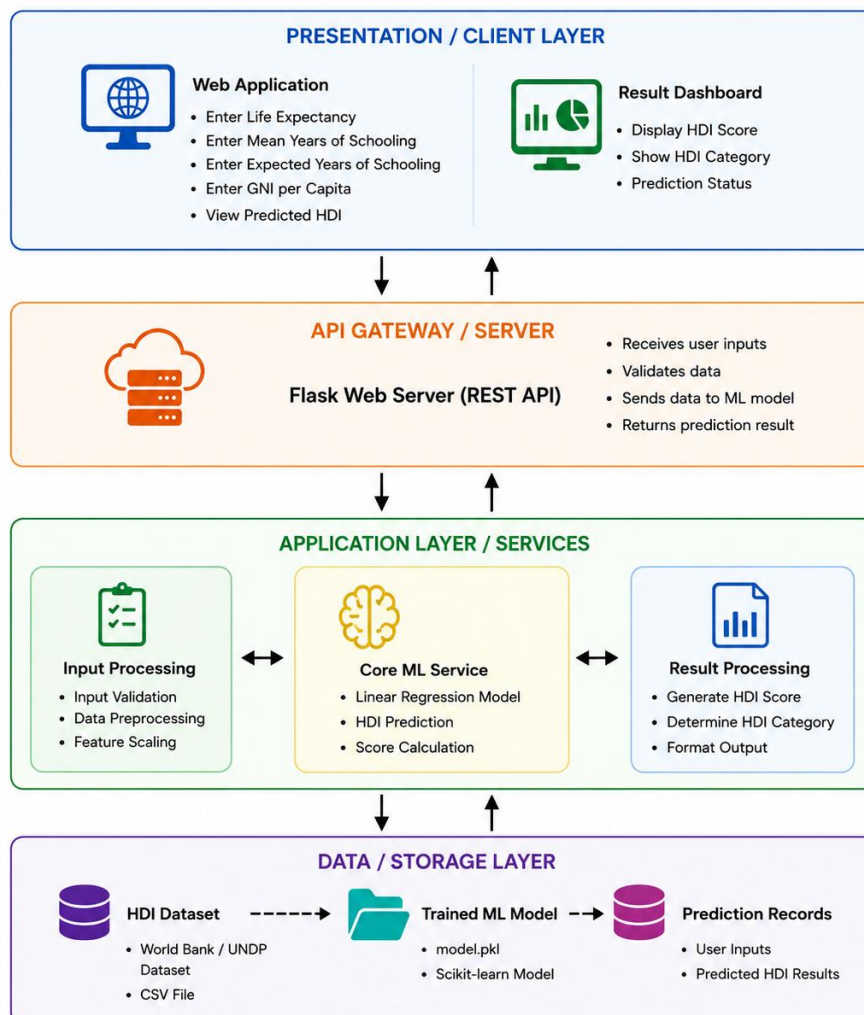
Solution Architecture

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

SOLUTION ARCHITECTURE DIAGRAM (HDI Prediction System)



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a web interface for users to enter development indicators and view the predicted HDI score and category.	HTML5, CSS3, JavaScript
API Gateway	Receives user requests, validates inputs, communicates with the machine learning model, and returns prediction results.	Python, Flask
Application Layer / Services	Validates user input, scales input features using the saved scaler, loads the trained Linear Regression model, predicts the HDI score, and determines the HDI category.	Python, Pandas, NumPy, Scikit-learn, Joblib
Data / Storage Layer	Stores the HDI dataset and trained machine learning model files used for prediction.	CSV Dataset, Pickle (.pkl), Joblib